

Fuel Property Models in FuelLib (`astm` branch): A Self-Contained Guide with All Equations

FuelLib documentation

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Abstract

This document explains, from first principles, every fuel property model implemented on the `astm` branch of FuelLib. It is written for a reader with no background in fuel chemistry or thermodynamics: each section first explains *what* the property is and *why* aviation fuel specifications regulate it, then gives the model equations exactly as implemented, the data the model relies on, and its verified accuracy. The models cover: composition bookkeeping, the Constantinou–Gani group contribution method, experimental anchoring of boiling/melting points and acentric factors, vapor pressure, density, heat capacities, latent heat, viscosity, surface tension, UNIFAC activity coefficients, and the six ASTM qualification properties — net heat of combustion, flash point, freezing point, Yield Sooting Index, Derived Cetane Number, and (via companion distillation models) the D86 distillation curve.

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1 How FuelLib represents a fuel

A real jet fuel contains thousands of distinct molecules. Nobody models them individually. Instead, a fuel is measured by two-dimensional gas chromatography (GC×GC), which sorts molecules into *bins* by chemical family (normal alkanes, branched “iso” alkanes, one-ring cycloalkanes, aromatics, ...) and by carbon number (how many carbon atoms each molecule has). A typical jet fuel needs 60–70 bins; each bin reports a weight percent.

FuelLib treats every bin as a single *pseudo-component*: one representative molecule (the *reference compound*, named in `fuelData/gcData/{fuel}_init.csv`) stands in for everything in the bin. All fuel properties are then computed from (i) per-bin constants determined once from the reference molecule’s structure, and (ii) the composition vector.

Composition bookkeeping. Let Y_i be mass fractions ($\sum_i Y_i = 1$), X_i mole fractions and ϕ_i liquid volume fractions. With molecular weights M_i (kg/mol) and liquid densities ρ_i (kg/m³):

$$\bar{M} = \left(\sum_i \frac{Y_i}{M_i} \right)^{-1}, \quad X_i = \frac{\bar{M} Y_i}{M_i}, \quad \phi_i = \frac{Y_i / \rho_i}{\sum_j Y_j / \rho_j}. \quad (1)$$

Different properties blend naturally on different bases: energies per unit mass blend by Y_i , phase equilibrium by X_i , and refinery-style indices (cetane) by ϕ_i .

2 The Constantinou–Gani group contribution method

Most reference compounds have no complete experimental datasheet, so their constants are *estimated from structure*. The idea of a group contribution (GC) method: chop the molecule into standard functional groups (CH₃, CH₂, aromatic CH, rings, ...), count them, and sum tabulated contributions. FuelLib uses the Constantinou–Gani (CG) method (1994/1995). With N_{ik} = number of first-order groups k in compound i and tabulated contributions c_k per property (file `gcmTableData/gcmTable.csv`):

$$M_i = \sum_k N_{ik} m_k \quad \text{molecular weight} \quad (2)$$

$$T_{c,i} = 181.128 \ln \left(\sum_k N_{ik} t_{c,k} \right) \quad \text{critical temperature (K)} \quad (3)$$

$$P_{c,i} = \left[\left(\sum_k N_{ik} p_{c,k} + 0.10022 \right)^{-2} + 1.3705 \right] \times 10^5 \quad \text{critical pressure (Pa)} \quad (4)$$

$$T_{b,i} = 204.359 \ln \left(\sum_k N_{ik} t_{b,k} \right) \quad \text{normal boiling point (K)} \quad (5)$$

$$T_{m,i} = 102.425 \ln \left(\sum_k N_{ik} t_{m,k} \right) \quad \text{melting point (K)} \quad (6)$$

$$\omega_i = 0.4085 \left[\ln \left(\sum_k N_{ik} w_k + 1.1507 \right) \right]^{1/0.5050} \quad \text{acentric factor} \quad (7)$$

$$V_{m,i}^{298} = \left(\sum_k N_{ik} v_k + 0.01211 \right) \times 10^{-3} \quad \text{molar liquid volume (m}^3\text{/mol)} \quad (8)$$

plus analogous sums for the ideal-gas enthalpy of formation $\Delta H_{f,i}$ (J/mol), the vaporization enthalpy at 298 K $\Delta H_{v,i}^{298}$, and heat-capacity polynomial coefficients. An *extended* table (`gcmExtendedTable.csv`) adds per-group atom counts (n_C , n_H), Ruzicka–Domalski liquid- C_p coefficients, and Alibakhshi flash-point contributions, all evaluated by the same $\sum_k N_{ik}(\cdot)_k$ pattern.

Where GC methods fail — and the fix. Group counting cannot see molecular *symmetry* or isomer detail. This matters most for the melting point (crystal packing) and for strongly branched molecules. Two measured examples: CG gives T_m of *n*-decane as 217 K versus the true 243.5 K; and indane’s T_b is over-predicted by 54 K. The branch therefore **anchors** T_b , T_m and ω to experimental values where they are known (`gcmTableData/property_anchors.csv`; provenance-tagged `nist` or homologous-series), keeping CG only as the fallback. For compounds with an anchored T_b but no measured ω , the acentric factor is re-derived from the Kesler–Lee closure so that the predicted vapor pressure equals 1 atm exactly at the anchored boiling point:

$$\omega = \frac{-\ln(P_c/101325) - f^{(0)}(T_b/T_c)}{f^{(1)}(T_b/T_c)}, \quad (9)$$

with $f^{(0)}, f^{(1)}$ the Lee–Kesler functions of Section 3.1. The closure is applied only when $T_b/T_c < 0.90$ and the result lies in $(0, 1.2)$ (outside that range it is numerically degenerate and CG’s ω is kept).

3 Per-compound temperature-dependent properties

3.1 Vapor pressure (Lee–Kesler)

The vapor pressure $P^{\text{sat}}(T)$ is the pressure at which a liquid boils at temperature T ; it controls distillation, flash point, and evaporation. FuelLib uses the Lee–Kesler corresponding-states form. With reduced temperature $T_r = T/T_c$:

$$f^{(0)} = 5.92714 - \frac{6.09648}{T_r} - 1.28862 \ln T_r + 0.169347 T_r^6, \quad (10)$$

$$f^{(1)} = 15.2518 - \frac{15.6875}{T_r} - 13.4721 \ln T_r + 0.43577 T_r^6, \quad (11)$$

$$P_i^{\text{sat}}(T) = P_{c,i} \exp(f^{(0)} + \omega_i f^{(1)}). \quad (12)$$

The acentric factor ω is precisely the knob that tilts the vapor pressure curve; that is why the anchoring strategy above treats it together with T_b .

3.2 Molar liquid volume and density (Rackett form)

$$V_{m,i}(T) = V_{m,i}^{298} z_i^{\phi_i(T)}, \quad z_i = 0.29056 - 0.08775 \omega_i, \quad (13)$$

$$\phi_i(T) = \begin{cases} (1 - T/T_{c,i})^{2/7} - (1 - 298/T_{c,i})^{2/7}, & T \leq T_{c,i}, \\ -(1 - 298/T_{c,i})^{2/7}, & T > T_{c,i}, \end{cases} \quad (14)$$

and the density is $\rho_i(T) = M_i/V_{m,i}(T)$. The mixture density is volume-additive: $\rho_{\text{mix}}(T) = \sum_i Y_i M_i/V_{m,i}(T)$ per unit mass — i.e. $1/\rho_{\text{mix}} = \sum_i Y_i/\rho_i$.

3.3 Latent heat of vaporization (Watson)

The energy to evaporate unit mass of liquid, temperature-corrected from the 298 K group value:

$$L_{v,i}(T) = L_{v,i}^{298} \left(\frac{1 - T/T_{c,i}}{1 - T_b,i/T_{c,i}} \right)^{0.38}, \quad L_{v,i}^{298} = \frac{\Delta H_{v,i}^{298}}{M_i}, \quad (15)$$

set to zero above T_c .

3.4 Heat capacities

Molar (used as both liquid and vapor proxy in the distillation models):

$$C_{p,i}(T) = C_{p,i}^{298} + B_i \theta + C_i \theta^2, \quad \theta = \frac{T - 298}{700}. \quad (16)$$

Liquid-phase specific heat via Ruzicka–Domalski group additivity (J/kg/K):

$$\frac{C_{p,L,i}(T)}{R} = A_i + B_i \frac{T}{100} + D_i \left(\frac{T}{100} \right)^2, \quad (17)$$

divided by M_i ; validated to $< 1\%$ against NIST for n -alkanes at 298 K, with a documented $+2\% \rightarrow +13\%$ drift from -10 to 160°C on Jet A.

3.5 Kinematic viscosity (Dutt)

Resistance to flow, in m^2/s . Dutt’s equation (temperatures in $^\circ\text{C}$):

$$\nu_i(T) = 10^{-6} \exp \left(-3.0171 + \frac{442.78 + 1.6452 (T_{b,i} - 273.15)}{(T - 273.15) + 239 - 0.19 (T_{b,i} - 273.15)} \right). \quad (18)$$

Mixtures blend by the Kendall–Monroe cube-root rule $\nu_{\text{mix}} = (\sum_i X_i \nu_i^{1/3})^3$ (an Arrhenius log-linear option also exists).

3.6 Surface tension (Brock–Bird)

With P_c in bar and $T_{br} = T_b/T_c$:

$$Q_i = 0.1196 \left(1 + \frac{T_{br,i} \ln(P_{c,i}/1.01325)}{1 - T_{br,i}} \right) - 0.279, \quad (19)$$

$$\sigma_i(T) = 10^{-3} P_{c,i}^{2/3} T_{c,i}^{1/3} Q_i (1 - T/T_{c,i})^{11/9} \quad [\text{N/m}]. \quad (20)$$

Known limitation: after the T_b/ω anchoring, Brock–Bird’s POSF-fuel accuracy degraded from $\sim 1\text{--}4\%$ to $\sim 4\text{--}13\%$ — the old agreement partly relied on cancellation between CG errors. A retune of Q is the recorded follow-up.

4 Mixture non-ideality: UNIFAC 2.0 activity coefficients

Raoult’s law ($P_i = X_i P_i^{\text{sat}}$) assumes molecules of different species interact identically. Real mixtures deviate; the correction factor is the activity coefficient γ_i : $P_i = \gamma_i X_i P_i^{\text{sat}}$. FuelLib implements classical UNIFAC with the UNIFAC 2.0 parameter matrix (Hayer et al. 2025): molecules are decomposed into 113 standardized subgroups with surface/volume parameters (Q_k, R_k), and a 54×54 main-group interaction matrix a_{mn} (K) enters through $\Psi_{mn} = \exp(-a_{mn}/T)$. The coefficient splits into a size/shape (“combinatorial”) part and an energetic (“residual”) part, $\ln \gamma_i = \ln \gamma_i^C + \ln \gamma_i^R$, with

$$\ln \gamma_i^C = 1 - V_i + \ln V_i - 5q_i \left(1 - \frac{V_i}{F_i} + \ln \frac{V_i}{F_i} \right), \quad V_i = \frac{r_i}{\sum_j X_j r_j}, \quad F_i = \frac{q_i}{\sum_j X_j q_j}, \quad (21)$$

where $r_i = \sum_k \nu_{ik} R_k$, $q_i = \sum_k \nu_{ik} Q_k$, and the residual part is the standard solution-of-groups expression $\ln \gamma_i^R = \sum_k \nu_{ik} (\ln \Gamma_k - \ln \Gamma_k^{(i)})$. For jet-range hydrocarbon mixtures γ_i stays within a

few percent of unity — which is why ideal-solution approximations elsewhere (flash point, freezing point) are acceptable.

Implementation note: at load time the 113-subgroup basis is compressed to the subgroups the fuel actually uses (7 for the POSF fuels). The compression is exact — unused subgroups carry zero counts and zero weight in every sum — and speeds up γ evaluation by $\sim 6.6\times$ end-to-end.

5 ASTM qualification properties

5.1 Net heat of combustion (ASTM D4809) — `heat_of_combustion`

The energy released burning 1 kg of fuel with the product water remaining as vapor (“lower heating value”, LHV) — the headline energy metric of any aviation fuel (spec: ≥ 42.8 MJ/kg). Computed by a Hess-cycle: burn each pseudo-component $C_aH_b + (a + b/4) O_2 \rightarrow a CO_2 + (b/2) H_2O(g)$,

$$LHV_i = -\frac{a_i \Delta H_f^{CO_2} + \frac{b_i}{2} \Delta H_f^{H_2O(g)} - \Delta H_{f,i}^{liq}}{10^6 M_i} \quad [MJ/kg], \quad (22)$$

with $\Delta H_f^{CO_2} = -393.51$ kJ/mol, $\Delta H_f^{H_2O(g)} = -241.83$ kJ/mol, and the liquid-phase formation enthalpy $\Delta H_{f,i}^{liq} = \Delta H_{f,i} - \Delta H_{v,i}^{298}$ (the CG value is for the ideal gas; D4809 burns liquid). The mixture blends by mass: $LHV = \sum_i Y_i LHV_i$. A guard derived from the group-name metadata rejects non-hydrocarbon fuels. Accuracy: Jet A predicted 43.38 MJ/kg vs. measured 43.17 (+0.5%).

5.2 Flash point (ASTM D93) — `flash_point`

The lowest temperature at which the vapor above the liquid can ignite — the fuel-handling safety property (Jet A spec: $\geq 38^\circ C$). Two-step model:

1. *Pure-compound flash points* from Alibakhshi’s correlation (AAD ≈ 5.8 K over 1533 compounds):

$$T_{f,i} = 12.14 + 0.73 T_{b,i} + \sum_k N_{ik} \phi_k. \quad (23)$$

2. *Mixture rule:* the ideal Liaw–Chiu criterion — the flash point is the temperature T at which the Le Chatelier sum of the components’ vapor contributions reaches unity:

$$\sum_i \frac{X_i P_i^{\text{sat}}(T)}{P_i^{\text{sat}}(T_{f,i})} = 1. \quad (24)$$

Each term is component i ’s fraction of its “own” flammable vapor pressure; activity coefficients are set to 1 (validated for HC–HC jet mixtures to $\sim 1^\circ C$ in the literature). Eq. (24) is solved by a fixed-iteration Newton method (the residual is monotone in T).

5.3 Freezing point (ASTM D5972) — `freeze_point`

The temperature at which the first crystal appears on cooling — the cold-operability property (Jet A spec: $\leq -40^\circ C$; Jet A-1 $\leq -47^\circ C$). Physics: the compound with the strongest tendency to crystallize sets the fuel’s freezing point. For each candidate j at mole fraction x_j , the branch solves Boehm et al.’s (2022) form of the solid–liquid equilibrium, by fixed-point iteration:

$$T^{(n+1)} = \frac{\Delta H_{\text{fus},j} + x_j \Delta c_{p,j} (T_{m,j} - T^{(n)})}{\Delta S_{\text{fus},j} + x_j \Delta c_{p,j} \ln(T^{(n)}/T_{m,j}) + \alpha \Delta S_{\text{mix}}}, \quad \Delta S_{\text{mix}} = -\frac{R}{x_j} [(1-x_j) \ln(1-x_j) + x_j \ln x_j], \quad (25)$$

and the fuel freezes at $\max_j T_j$. Dilution enters through $\Delta S_{\text{mix}} \approx -R \ln x_j$ (small x_j), which is exactly the classical ideal-solubility depression term — hence $\alpha = 1$ *when physical fusion data are used* (the historical $\alpha = 0.25$ was only ever valid paired with the crude Walden approximation; the pair’s errors partially cancelled).

Fusion data: $\Delta S_{\text{fus},j} = A+B(n_{C,j}-C_0)$ per chemical family (`gcmTableData/fusion_families.csv`); the n -alkane line is anchored to NIST data (e.g. n -C12: 140 J/mol/K — Walden’s universal 56.5 was off by $2.5\times$), $\Delta H_{\text{fus}} = \Delta S_{\text{fus}} T_m$, and $\Delta c_p \approx -0.35 C_{p,L}(298)$.

Verified accuracy (after T_m anchoring): pure n -alkanes within 0.6 K of NIST; POSF fuels within -7.6 to -0.1 K of the Edwards references.

5.4 Yield Sooting Index — `ysi`

A measured, dimensionless ranking of a molecule’s tendency to form soot (unified scale: benzene = 100, n -hexane = 30), from the Yale YSI database. Per-bin values live in `gcmTableData/das_2018_ysi.csv` with a provenance tag per row (measured / extrapolated-in-family / cross-filled); of the 89 skeleton bins, 35 are direct measurements. The mixture blends linearly in mole fraction:

$$\text{YSI} = \sum_i X_i \text{YSI}_i, \quad \sigma_{\text{YSI}} = \left(\sum_i (X_i \sigma_i)^2 \right)^{1/2}, \quad (26)$$

where per-bin uncertainties σ_i are the tabulated measurement errors, inflated ($\max(2\times, 15\%)$) for non-measured rows; unresolved bins are filled with the family mean (flagged, $\sigma \geq 30\%$). The uncertainty propagation assumes independent errors and is therefore a lower bound — family-extrapolated errors are correlated.

5.5 Derived Cetane Number (ASTM D6890) — `dcn`

DCN measures autoignition quality in a constant-volume combustion chamber (IQT): high DCN ignites easily (long n -alkanes, DCN of n -hexadecane $\equiv 100$), low DCN resists ignition (aromatics < 10 ; heavily branched alkanes ~ 15 – 25 , with heptamethylnonane $\equiv 15$ defining the scale’s low end). DCN discriminates *branching* more sharply than any other property here — e.g. n -C12 ≈ 74 vs. pentamethylheptane ≈ 17 .

$$\text{DCN} = \sum_i \phi_i \text{DCN}_i \quad (\text{linear in liquid } \textit{volume} \text{ fraction}), \quad (27)$$

with per-bin values in `gcmTableData/dcn.csv`: 16 literature seed anchors (flagged for verification against the NREL Compendium of Experimental Cetane Numbers) plus family fits and offset rules, every row provenance-tagged, and σ via the same independent-error propagation as YSI. The generic isoparaffin bins use a *calibrated* multi-branched archetype (offset -25 from the n -alkane line); fuel-specific bins can override it, as done for the ATJ fuel below.

Verified accuracy (published NJFCP references):

fuel	NJFCP id	measured	predicted	Δ
posf10264 (JP-8)	A-1	48.8	51.6	+2.8
posf10325 (Jet A)	A-2	48.3	51.4	+3.1
posf10289 (JP-5)	A-3	39.2	50.8	+11.6 ¹
posf11498 (ATJ)	C-1	17.1	19.7	+2.6
hefa-came (HEFA)	—	~ 55 – 60	58.2	in band

The C-1 row deserves emphasis: with the default (lightly-branched) isoparaffin archetype the prediction was +39 off. Re-mapping the fuel’s two dominant bins to the true ATJ isomers (2,2,4,6,6-pentamethylheptane and heptamethylnonane) in the per-fuel data files brought it to +2.6 *without touching the model* — the single clearest demonstration that pseudo-component fidelity, not correlation quality, limits accuracy for synthetic fuels.

5.6 Distillation curve (ASTM D86)

The D86 curve (temperature vs. volume recovered in a standardized batch distillation) is modeled by the companion distillation modules (three-stage still: pot bubble-point + energy balance, column-neck flash, ε -NTU condenser, thermometer lag, and an inverse-model feed-forward heater controller). It is documented separately (`docs/d86_simulation_algorithm.tex` on its home branch) and is out of scope here; the property models above (vapor pressure, activity, latent heat, C_p) are exactly its inputs.

6 Uncertainty, limitations, and honest caveats

- **Pseudo-component fidelity is the leading error source** for synthetic fuels: a bin’s reference compound must actually resemble the fuel’s molecules (see the ATJ case above). Audit reference compounds before trusting any property of a new SAF.
- **Melting points:** anchored for 36/89 bins; un-anchored bins keep CG T_m , which cannot see crystal symmetry. Freezing-point predictions for fuels dominated by un-anchored bins inherit that error.
- **Surface tension and mid-range vapor pressure** lost some accuracy when T_b/ω anchoring removed error cancellations they had been co-tuned with (ST on POSF fuels now 4–13%). Recorded follow-up: retune Brock–Bird’s Q .
- **DCN seeds** await line-by-line verification against the NREL compendium; every derived value carries a provenance tag and an inflated uncertainty.
- **Uncertainty propagation** (YSI, DCN) assumes independent per-bin errors; family-extrapolated errors are correlated, so reported σ are lower bounds.
- **Blending rules** are first-order (linear in the appropriate basis; Kendall–Monroe for viscosity). Known second-order effects (cetane blending non-linearity, sooting synergies) are not modeled.

¹JP-5 is cycloparaffin-rich; the cycloalkane DCN family curves rest on only two measured seeds. More measured cycloalkane DCNs, not code, tighten this.

Data files at a glance

file	contents
<code>gcmTableData/gcmTable.csv</code>	CG first/second-order group contributions
<code>gcmTableData/gcmExtendedTable.csv</code>	atom counts, RD C_p , Alibakhshi ϕ_k
<code>gcmTableData/property_anchors.csv</code>	experimental $T_b/T_m/\omega$ anchors
<code>gcmTableData/fusion_families.csv</code>	family ΔS_{fus} correlations
<code>gcmTableData/das_2018_ysi.csv</code>	per-bin YSI + uncertainty + provenance
<code>gcmTableData/dcn.csv</code>	per-bin DCN + uncertainty + provenance
<code>unifacTableData/*</code>	UNIFAC 2.0 subgroup R, Q and a_{mn}
<code>fuelData/gcData/{fuel}_init.csv</code>	bins, reference compounds, wt%
<code>fuelData/groupDecompositionData/*</code>	CG group counts per bin
<code>fuelData/unifacDecomposition/*</code>	UNIFAC subgroup counts per bin